

README

E2E proof — multi-provider + stale model-allowlist audit (§11.4.107 / §11.4.6 / §11.4.150)

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Multi-provider breadth — PROVEN

Two distinct real providers translate the same source EN→Serbian Cyrillic end-to-end:

- **DeepSeek** (deepseek-chat) — proven across all 5 input formats (see `e2e_format_matrix_*`).
- **Mistral** (mistral-small-latest) — PASS, 247 Cyrillic chars: ****Врана и крчаг**** ... Једна жедна врана пронашла је крчаг у којем је на дну било мало воде...

Real bug fixed this round (proven)

`cmd/unified-translator resolveProviderAPIKey env-key map` was MISSING `zhipu`, so `ZHIPU_API_KEY` never satisfied the gate ('API key required for provider=zhipu'). Added `"zhipu": "ZHIPU_API_KEY"`. PROOF: after the fix, `zhipu` passes the gate and reaches the real Zhipu API (it now returns a model-name error from the live API instead of the local gate error — gate fixed).

FLAGGED — systemic stale provider model-allowlists (real, NOT fixed; research-gated §11.4.150)

The per-provider valid-model allowlists in `pkg/translator/llm` reject CURRENT models and only accept DEPRECATED ones (a moving target; needs per-provider current-model research before changing):

Provider	Allowlist accepts (stale)	Current models it REJECTS	Evidence
gemini	gemini-pro, gemini-pro-vision	gemini-2.0-flash, gemini-1.5-*	"model 'gemini-2.0-flash' is not valid... Valid models: [gemini-pro gemini-pro-vision]"
groq	llama-3.1-70b-versatile, llama-3.1-8b-instant, mixtral-8x7b-32768	llama-3.3-70b-versatile	"Valid models: [llama-3.1-70b-versatile llama-3.1-8b-instant mixtral-8x7b-32768]"
zhipu	glm-4, glm-3-turbo	glm-4-flash, glm-4-plus, glm-4-air	factory allows glm-4 but the LIVE API returns code 1211 "模型不存在" (glm-4 no longer exists)

DeepSeek + Mistral allowlists are current (both translate successfully).